PROJECT CODEBOOK

Employee Portal Application

This PDF is a reviewer-friendly code walkthrough of the Employee Portal assessment project. It includes a short overview of the solution and then presents the source code in two grouped sections: frontend and backend. Each file is shown under its own header so the interviewer can read the implementation directly from the document.

PRIMARY REPOSITORY LINK

https://github.com/kezama1231/EmployeePortalApp

Project Overview

The application is an employee management portal that supports core CRUD operations for employee records. Users can browse employees by department, view employee details, create new employees, update existing records, and delete employees. The frontend communicates with a Spring Boot REST API, which persists data using JPA and H2.

https://github.com/kezama1231/EmployeePortalApp

Technology Used

- React + Vite
- PrimeReact
- Axios
- React Router
- Spring Boot
- Spring Web
- Spring Data JPA
- H2 Database
- Flyway
- MapStruct

Frontend Files

- employee-portal-frontend/package.json
- employee-portal-frontend/index.html
- employee-portal-frontend/vite.config.js
- employee-portal-frontend/eslint.config.js

Backend Files

- EmployeePortalApp/pom.xml
- EmployeePortalApp/src/main/resources/application.properties

- employee-portal-frontend/src/main.jsx
- employee-portal-frontend/src/App.jsx
- employee-portal-frontend/src/App.css
- employee-portal-frontend/src/index.css
- employee-portal-frontend/src/api/employeeApi.js
- employee-portal-frontend/src/component/DepartmentFilter.jsx
- employee-portal-frontend/src/component/EmployeeForm.jsx
- employee-portal-frontend/src/component/EmployeeTable.jsx
- employee-portal-frontend/src/component/Layout.jsx
- employee-portal-frontend/src/pages/CreateEmployeePage.jsx
- employee-portal-frontend/src/pages/EditEmployeePage.jsx
- employee-portal-frontend/src/pages/EmployeeDetailsPage.jsx
- employee-portal-frontend/src/pages/EmployeeListPage.jsx
- employee-portal-frontend/src/pages/NotFoundPage.jsx
- EmployeePortalApp/src/main/resources/db/migration/V1__create_employee_table.sql
- EmployeePortalApp/src/main/resources/db/migration/V2__insert_employee_seed_data.sql
- EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/EmployeePortalAppApplication.java
- EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/config/AuditConfig.java
- EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/config/AuditorAwareImpl.java
- EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/controller/EmployeeController.java
- EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/dto/EmployeeCreateRequest.java
- EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/dto/EmployeeUpdateRequest.java
- EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/dto/EmployeeResponse.java
- EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/entity/Employee.java
- EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/exception/EmployeeNotFoundException.java
- EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/exception/GlobalExceptionHandler.java
- EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/mapper/EmployeeMapper.java
- EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/repository/EmployeeRepository.java
- EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/service/EmployeeService.java
- EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/service/impl/EmployeeServiceImpl.java
- EmployeePortalApp/src/test/java/com/interview/EmployeePortalApp/EmployeePortalAppA

pplicationTests.java

SECTION 1

Frontend

This section contains the React application files used to render the employee portal UI, manage routing, make API requests, and style the user experience.

```
https://github.com/kezama1231/EmployeePortalApp
```

FRONTEND FILE

package.json

employee-portal-frontend/package.json

JSON

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```
{
  "name": "employee-portal-frontend",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "start" : "vite",
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
    "preview": "vite preview"
  },
  "dependencies": {
    "axios": "^1.13.6",
    "primeicons": "^7.0.0",
    "primereact": "^10.9.7",
    "react": "^19.2.4",
    "react-dom": "^19.2.4",
    "react-router-dom": "^7.13.1"
  },
  "devDependencies": {
    "@eslint/js": "^9.39.4",
    "@types/react": "^19.2.14",
    "@types/react-dom": "^19.2.3",
    "@vitejs/plugin-react": "^6.0.0",
    "eslint": "^9.39.4",
    "eslint-plugin-react-hooks": "^7.0.1",
    "eslint-plugin-react-refresh": "^0.5.2",
    "globals": "^17.4.0",
    "vite": "^8.0.0"
  }
}
```

FRONTEND FILE

index.html

employee-portal-frontend/index.html

HTML

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```html
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/favicon.svg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>employee-portal-frontend</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

FRONTEND FILE

vite.config.js

employee-portal-frontend/vite.config.js

JavaScript

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```
import { defineConfig } from 'vite'
import react from '@vitejs/plugin-react'

// https://vite.dev/config/
export default defineConfig({
  plugins: [react()],
})
```

FRONTEND FILE

eslint.config.js

employee-portal-frontend/eslint.config.js

JavaScript

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```
import js from '@eslint/js'
import globals from 'globals'
import reactHooks from 'eslint-plugin-react-hooks'
import reactRefresh from 'eslint-plugin-react-refresh'
import { defineConfig, globalIgnores } from 'eslint/config'

export default defineConfig([
  globalIgnores(['dist']),
  {
    files: ['**/*.{js,jsx}'],
    extends: [
      js.configs.recommended,
      reactHooks.configs.flat.recommended,
      reactRefresh.configs.vite,
    ],
    languageOptions: {
      ecmaVersion: 2020,
      globals: globals.browser,
      parserOptions: {
        ecmaVersion: 'latest',
        ecmaFeatures: { jsx: true },
        sourceType: 'module',
      },
    },
    rules: {
      'no-unused-vars': ['error', { varsIgnorePattern: '^[A-Z_]' }],
    },
  },
])
```

FRONTEND FILE

main.jsx

employee-portal-frontend/src/main.jsx

React JSX

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import { BrowserRouter } from 'react-router-dom'
import './index.css'
import App from './App.jsx'

import 'primereact/resources/themes/lara-light-cyan/theme.css';
import 'primeicons/primeicons.css';

createRoot(document.getElementById('root')).render(
  <StrictMode>
    <BrowserRouter>
      <App />
    </BrowserRouter>
  </StrictMode>,
)
```

FRONTEND FILE

App.jsx

employee-portal-frontend/src/App.jsx

React JSX

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```
import { Routes, Route } from 'react-router-dom';
import Layout from './component/Layout';
import './App.css';
import EmployeeListPage from './pages/EmployeeListPage';
import EmployeeDetailsPage from './pages/EmployeeDetailsPage';
import CreateEmployeePage from './pages/CreateEmployeePage';
import EditEmployeePage from './pages/EditEmployeePage';
import NotFoundPage from './pages/NotFoundPage';

function App() {

  return (
    <Routes>
      <Route path="/" element={<Layout />}>
        <Route index element={<EmployeeListPage/>} />
        <Route path="employees/create" element={<CreateEmployeePage/>} />
        <Route path="employees/:id" element={<EmployeeDetailsPage />} />
        <Route path="employees/:id/edit" element={<EditEmployeePage/>} />
        <Route path="*" element={<NotFoundPage />} />
      </Route>
    </Routes>
  )
}

export default App
```

FRONTEND FILE

App.css

employee-portal-frontend/src/App.css

CSS

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```
body {
  margin: 0;
  font-family: "Aptos", "Segoe UI", sans-serif;
  color: var(--text-main);
}

.app-shell {
  min-height: 100vh;
  padding: 28px 18px 40px;
}

.page-container {
  max-width: 1200px;
  margin: 0 auto;
  padding: 0 0 24px;
}

.app-title {
  font-size: 1.15rem;
  font-weight: 700;
  color: var(--text-main);
}

.app-subtitle {
  color: var(--text-soft);
  font-size: 0.82rem;
}

.brand-block {
  display: flex;
  align-items: center;
  gap: 14px;
}

.brand-badge {
  width: 42px;
  height: 42px;
  border-radius: 14px;
  display: flex;
  align-items: center;
  justify-content: center;
  background: linear-gradient(135deg, var(--brand-primary), var(--brand-primary-dark));
  color: #fff;
  font-size: 0.95rem;
```

```
  font-weight: 700;
  letter-spacing: 0.08em;
}

.nav-actions {
  display: flex;
  gap: 12px;
  align-items: center;
  flex-wrap: wrap;
}

.nav-link {
  text-decoration: none;
}

.portal-menubar {
  border: 1px solid var(--brand-border);
  border-radius: 24px;
  padding: 0.9rem 1.15rem;
  background: rgba(255, 255, 255, 0.96);
  box-shadow: 0 16px 40px rgba(17, 24, 39, 0.06);
}

.content-stack {
  display: grid;
  gap: 24px;
  margin-top: 24px;
}

.page-card {
  border: 0;
  border-radius: 26px;
  box-shadow: 0 18px 44px rgba(17, 24, 39, 0.06);
}

.page-card .p-card-body {
  padding: 28px;
}

.page-card .p-card-title {
  margin-bottom: 1.2rem;
  color: var(--text-main);
  font-size: 1.55rem;
  letter-spacing: -0.02em;
}

.filter-row {
  margin-bottom: 1.5rem;
  max-width: 320px;
}

.field-label {
  display: block;
  margin-bottom: 0.55rem;
  font-weight: 600;
  color: var(--text-soft);
}

.filter-dropdown,
.filter-row .p-dropdown {
  width: 100%;
```

```
}

.table-actions {
  display: flex;
  gap: 8px;
  flex-wrap: wrap;
}

.table-actions-compact .p-button {
  min-width: 92px;
  justify-content: center;
}

.action-btn {
  font-weight: 600;
}

.action-btn-view {
  color: var(--brand-primary);
  border-color: #cfd4dc;
  background: #fff;
}

.action-btn-view:hover {
  background: var(--brand-surface-muted);
  border-color: #bfc7d2;
}

.action-btn-edit {
  background: var(--brand-primary);
  border-color: var(--brand-primary);
  color: #fff;
}

.action-btn-edit:hover {
  background: var(--brand-primary-dark);
  border-color: var(--brand-primary-dark);
}

.action-btn-delete {
  background: #fff;
  color: #475467;
  border-color: #d1d5db;
}

.action-btn-delete:hover {
  background: #f3f4f6;
  color: #1f2937;
  border-color: #c5ccd6;
}

.employee-form {
  display: flex;
  flex-direction: column;
  gap: 18px;
  max-width: 560px;
}

.field-block {
  display: flex;
  flex-direction: column;
```

```
  gap: 8px;
}

.field-block label {
  font-weight: 600;
  color: var(--text-soft);
}

.employee-form .p-inputtext,
.employee-form .p-dropdown {
  width: 100%;
}

.page-actions {
  display: flex;
  gap: 12px;
  margin-top: 24px;
  flex-wrap: wrap;
}

.centered-block {
  display: flex;
  justify-content: center;
  padding: 40px 0;
}

.message-block {
  margin-bottom: 16px;
}

.detail-grid {
  display: grid;
  grid-template-columns: repeat(2, minmax(0, 1fr));
  gap: 14px;
}

.detail-grid > div {
  padding: 16px 18px;
  border-radius: 18px;
  background: var(--brand-surface);
  border: 1px solid var(--brand-border);
  color: var(--text-soft);
}

.detail-grid strong {
  color: var(--text-main);
}

.muted-text {
  color: var(--text-soft);
}

.employee-table {
  border: 1px solid var(--brand-border);
  border-radius: 20px;
  overflow: hidden;
}

.employee-table .p-datatable-thead > tr > th {
  background: var(--brand-surface-muted);
  color: var(--text-main);
```

```
  border-color: var(--brand-border);
  font-weight: 700;
}

.employee-table .p-datatable-tbody > tr > td {
  border-color: #e5e7eb;
}

.employee-table .p-datatable-tbody > tr:hover {
  background: #f9fafb;
}

.p-card {
  background: rgba(255, 255, 255, 0.95);
}

.p-button {
  border-radius: 12px;
}

.nav-actions .p-button {
  color: var(--brand-primary);
}

.nav-actions .p-button.p-button-outlined {
  border-color: var(--brand-primary);
}

.nav-actions .p-button.p-button-outlined:hover {
  background: var(--brand-primary);
  border-color: var(--brand-primary);
  color: #fff;
}

.p-inputtext,
.p-dropdown {
  border-radius: 14px;
  border-color: #d1d5db;
  background: #fff;
}

.p-inputtext:enabled:focus,
.p-dropdown:not(.p-disabled).p-focus {
  border-color: var(--brand-primary);
  box-shadow: 0 0 0 0.15rem rgba(51, 65, 85, 0.12);
}

@media (max-width: 960px) {
  .detail-grid {
    grid-template-columns: 1fr;
  }
}

@media (max-width: 640px) {
  .app-shell {
    padding-inline: 12px;
  }

  .portal-menubar {
    padding: 0.8rem;
  }
```

```
  .page-card .p-card-body {
    padding: 22px;
  }

  .table-actions,
  .page-actions {
    flex-direction: column;
    align-items: stretch;
  }

  .table-actions a,
  .page-actions a,
  .table-actions .p-button,
  .page-actions .p-button {
    width: 100%;
  }
}
```

FRONTEND FILE

index.css

employee-portal-frontend/src/index.css

CSS

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```
:root {
  font-family: "Aptos", "Segoe UI", sans-serif;
  line-height: 1.5;
  font-weight: 400;
  color: #111827;
  background: #f3f4f6;
  --brand-primary: #334155;
  --brand-primary-dark: #1f2937;
  --brand-primary-soft: #eef2f7;
  --brand-surface: #ffffff;
  --brand-surface-muted: #f9fafb;
  --brand-border: #d1d5db;
  --text-main: #111827;
  --text-soft: #4b5563;
  font-synthesis: none;
  text-rendering: optimizeLegibility;
  -webkit-font-smoothing: antialiased;
  -moz-osx-font-smoothing: grayscale;
}

* {
  box-sizing: border-box;
}

body {
  margin: 0;
  min-width: 320px;
  min-height: 100vh;
  background:
    radial-gradient(circle at top, rgba(51, 65, 85, 0.05), transparent 22%),
    linear-gradient(180deg, #f9fafb 0%, #eef2f7 100%);
}

#root {
  min-height: 100vh;
}

h1,
h2,
h3,
p {
  margin: 0;
}
```

```
h2 {
  color: var(--text-main);
}
```

FRONTEND FILE

employeeApi.js

employee-portal-frontend/src/api/employeeApi.js

JavaScript

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```javascript
import axios from "axios";

const api = axios.create({
    baseURL: 'http://localhost:8080/api/employees',
    headers: {
        'Content-Type': 'application/json',
    },
});

export const EmployeeService = {
    getEmployeesByDepartment: async (department) => {
        const response = await api.get(
            '', { params: {department}}
        );
        return response.data;
    },
    getEmployeeById: async (id) => {
        const response = await api.get(`/${id}`);
        return response.data;
    },
    createEmployee: async (employeeData) => {
        const response = await api.post('', employeeData);
        return response.data;
    },
    updateEmployee: async (id, employeeData) => {
        const response = await api.put(`/${id}`, employeeData);
        return response.data;
    },
    deleteEmployee: async (id) => {
        await api.delete(`/${id}`);
    }
}
```

FRONTEND FILE

DepartmentFilter.jsx

employee-portal-frontend/src/component/DepartmentFilter.jsx

React JSX

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```
import { Dropdown } from 'primereact/dropdown';

const departmentOptions = [
    { label: 'HR', value: 'HR' },
    { label: 'IT', value: 'IT' },
    { label: 'Finance', value: 'Finance' },
    { label: 'Operations', value: 'Operations' },
]

export default function DepartmentFilter({ department, onDepartmentChange }) {
    return (
        <div className="filter-row">
            <label htmlFor="departmentFilter" className="field-label">Department</label>
            <Dropdown
                inputId="departmentFilter"
                value={department}
                onChange={(e) => onDepartmentChange(e.value)}
                options={departmentOptions}
                optionLabel="label"
                optionValue="value"
                placeholder="Select department"
                className="filter-dropdown"
            />
        </div>
    );
}
```

FRONTEND FILE

EmployeeForm.jsx

employee-portal-frontend/src/component/EmployeeForm.jsx

React JSX

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```
import { Dropdown } from "primereact/dropdown";
import { InputText } from "primereact/inputtext";
import { Message } from "primereact/message";
import { useState, useEffect } from "react";
import { Button } from 'primereact/button';

const departmentOptions = [
    { label: 'HR', value: 'HR' },
    { label: 'IT', value: 'IT' },
    { label: 'Finance', value: 'Finance' },
    { label: 'Operations', value: 'Operations' },
];

const defaultFormValues = {
    name: '',
    contactNumber: '',
    department: '',
    email: '',
    jobTitle: '',
};

export default function EmployeeForm({ initialValues = defaultFormValues, onSubmit, submitLabel }) {

    const [formData, setFormData] = useState(defaultFormValues);
    const [error, setError] = useState('');

    useEffect(() => {
        setFormData(initialValues || defaultFormValues);
    }, [initialValues])

    const handleChange = (field, value) => {
        setFormData((previous) => ({
            ...previous,
            [field]: value,
        }));
    }

    const handleSubmit = async (event) => {
        event.preventDefault();
        setError('');

        try {
            await onSubmit(formData);
        } catch (err) {
```

```
            console.log(err);
            setError('Failed to submit employee form.');
        }
    };

    return (
        <form onSubmit={handleSubmit} className="employee-form">
            {error && <Message severity="error" text={error} />}

            <div className="field-block">
                <label htmlFor="name">Name</label>
                <InputText
                    id="name"
                    value={formData.name}
                    onChange={(e) => handleChange("name", e.target.value)}
                    required
                />
            </div>

            <div className="field-block">
                <label htmlFor="contactNumber">Contact Number</label>
                <InputText
                    id="contactNumber"
                    value={formData.contactNumber}
                    onChange={(e) => handleChange("contactNumber", e.target.value)}
                    required
                />
            </div>

            <div className="field-block">
                <label htmlFor="department">Department</label>
                <Dropdown
                    inputId="department"
                    value={formData.department}
                    onChange={(e) => handleChange("department", e.target.value)}
                    options={departmentOptions}
                    optionLabel="label"
                    optionValue="value"
                    placeholder="Select Department"
                />
            </div>

            <div className="field-block">
                <label htmlFor="email">Email</label>
                <InputText
                    id="email"
                    value={formData.email}
                    onChange={(e) => handleChange('email', e.target.value)}
                    required
                />
            </div>

            <div className="field-block">
                <label htmlFor="jobTitle">Job Title</label>
                <InputText
                    id="jobTitle"
                    value={formData.jobTitle}
                    onChange={(e) => handleChange('jobTitle', e.target.value)}
                    required
                />
            </div>
```

```
            <div className="page-actions">
                <Button type="submit" label={submitLabel} />
            </div>

        </form>
    )
}
```

FRONTEND FILE

EmployeeTable.jsx

employee-portal-frontend/src/component/EmployeeTable.jsx

React JSX

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```
import { Link } from 'react-router-dom';
import { DataTable } from 'primereact/datatable';
import { Column } from 'primereact/column';
import { Button } from 'primereact/button';

export default function EmployeeTable({ employees, onDelete }) {
    const actionBodyTemplate = (rowData) => {
        return (
            <div className="table-actions table-actions-compact">
                <Link to={`/employees/${rowData.employeeId}`}>
                    <Button
                        label={"View"}
                        icon="pi pi-eye"
                        size="small"
                        outlined
                        className="action-btn action-btn-view"
                    />
                </Link>
                <Link to={`/employees/${rowData.employeeId}/edit`}>
                    <Button
                        label={"Edit"}
                        icon="pi pi-pencil"
                        size="small"
                        className="action-btn action-btn-edit"
                    />
                </Link>
                <Button
                    label="Delete"
                    icon="pi pi-trash"
                    size="small"
                    severity="danger"
                    className="action-btn action-btn-delete"
                    onClick={() => onDelete(rowData.employeeId)}
                />
            </div>

        )
    }

    return (
        <DataTable
            className="employee-table"
            value={employees || []}
            stripedRows
```

```
            rows={15}
            paginator
            emptyMessage={"No employees  found for this department."}
            tableStyle={{ minWidth: '50rem' }}
        >
            <Column field="employeeId" header="ID" />
            <Column field="name" header="Name" sortable />
            <Column field="contactNumber" header="Contact Number" />
            <Column field="email" header="Email" />
            <Column field="jobTitle" header="Job Title" />
            <Column field="department" header="Department" />
            <Column header="Actions" body={actionBodyTemplate} />
        </DataTable>
    );
}
```

FRONTEND FILE

Layout.jsx

employee-portal-frontend/src/component/Layout.jsx

React JSX

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```
import { Button } from "primereact/button";
import { Menubar } from "primereact/menubar";
import { Outlet, Link } from "react-router-dom";

export default function Layout() {
    const startContent = (
        <div className="brand-block">
            <span className="brand-badge">EP</span>
            <div>
                <div className="app-title">Employee Portal</div>
                <div className="app-subtitle">Professional employee management</div>
            </div>
        </div>
    );

    const endContent = (
        <div className="nav-actions">
            <Link to="/" className="nav-link">
                <Button label={"Employees"} icon="pi pi-users" text />
            </Link>
            <Link to="/employees/create" className="nav-link">
                <Button label={"Create Employee"} icon="pi pi-plus" outlined />
            </Link>
        </div>
    );

    return (
        <div className="app-shell">
            <div className="page-container">
                <Menubar start={startContent} end={endContent} className="portal-menubar" />
                <div className="content-stack">
                    <Outlet />
                </div>
            </div>
        </div>
    );
}
```

FRONTEND FILE

CreateEmployeePage.jsx

employee-portal-frontend/src/pages/CreateEmployeePage.jsx

React JSX

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```
import { useNavigate } from "react-router-dom";
import EmployeeForm from "../component/EmployeeForm";
import { Card } from "primereact/card";
import { EmployeeService } from "../api/employeeApi";


export default function CreateEmployeePage() {

    const navigate = useNavigate();

    const handleCreateEmployee = async (formData) => {
        await EmployeeService.createEmployee(formData);
        navigate('/');
    };

    return (
        <Card title="Create Employee" className="page-card">
            <EmployeeForm
                onSubmit={handleCreateEmployee}
                submitLabel="Create Employee"
            />
        </Card>
    );
}
```

FRONTEND FILE

EditEmployeePage.jsx

employee-portal-frontend/src/pages/EditEmployeePage.jsx

React JSX

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```
import { useNavigate, useParams } from "react-router-dom";
import EmployeeForm from "../component/EmployeeForm";
import { useEffect, useState } from "react";
import { EmployeeService } from "../api/employeeApi";
import { Message } from "primereact/message";
import { ProgressSpinner } from "primereact/progressspinner";
import { Card } from "primereact/card";


export default function EditEmployeePage() {

    const { id } = useParams();
    const navigate = useNavigate();

    const [loading, setLoading] = useState(true);
    const [error, setError] = useState('');
    const [initialData, setInitialData] = useState(null);

    useEffect(() => {
        const fetchData = async () => {
            setLoading(true);
            setError('');
            try {
                const data = await EmployeeService.getEmployeeById(id);
                setInitialData({
                    name: data.name,
                    contactNumber: data.contactNumber,
                    department: data.department,
                    email: data.email,
                    jobTitle: data.jobTitle,
                });
            } catch (err) {
                setError('Failed to load employee.');
            } finally {
                setLoading(false);
            }
        };
        fetchData();
    }, [id]);

    const handleUpdateEmployee = async (formData) => {
        await EmployeeService.updateEmployee(id, formData);
        navigate(`/`);
    }
```

```
    if (loading) {
        return (
            <div className="centered-block">
                <ProgressSpinner />
            </div>
        );
    }

    if (error) {
        return <Message severity="error" text={error} />
    }


    return (
        <Card title="Edit Employee" className="page-card">
            <EmployeeForm
                initialValues={initialData}
                onSubmit={handleUpdateEmployee}
                submitLabel="Update Employee"
            />
        </Card>
    );
}
```

FRONTEND FILE

EmployeeDetailsPage.jsx

employee-portal-frontend/src/pages/EmployeeDetailsPage.jsx

React JSX

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```
import { useEffect, useState } from 'react';
import { Link, useNavigate, useParams } from 'react-router-dom';
import { Card } from 'primereact/card';
import { Button } from 'primereact/button';
import { Message } from 'primereact/message';
import { ProgressSpinner } from 'primereact/progressspinner';
import { ConfirmDialog, confirmDialog } from 'primereact/confirmdialog';
import { EmployeeService } from "../api/employeeApi";

export default function EmployeeDetailsPage() {

    const { id } = useParams();
    const navigate = useNavigate();
    const [employee, setEmployee] = useState(null);
    const [loading, setLoading] = useState(true);
    const [error, setError] = useState('');

    useEffect(() => {
        const fetchData = async () => {
            try {
                setLoading(true);
                setError('')
                const data = await EmployeeService.getEmployeeById(id);
                setEmployee(data);
            } catch (err) {
                console.log(err);
                setError('Employee not found.');
            } finally {
                setLoading(false);
            }
        }
        fetchData();
    }, [id])

    const handleDelete = () => {
        confirmDialog({
            message: "Before that!! Are you sure to delete ?",
            header: "Delete Employee",
            icon: "pi pi-exclamation-triangle",
            acceptClassName: "p-button-danger",
            accept: async () => {
                try {
                    await EmployeeService.deleteEmployee(id);
                    navigate("/");
```

```
                } catch (err) {
                    setError('Failed to delete employee.');
                }
            },
        });
    }

    if (loading) {
        return (
            <div className="centered-block">
                <ProgressSpinner />
            </div>

        )
    }

    if (error) {
        return (
            <Message severity="error" text={error} />
        )
    }

    return (
        <Card title={"Employee Details"} className="page-card">
            <ConfirmDialog />
            <div className="detail-grid">
                <div><strong>ID:</strong> {employee.employeeId}</div>
                <div><strong>Name:</strong> {employee.name}</div>
                <div><strong>Contact Number:</strong> {employee.contactNumber}</div>
                <div><strong>Email:</strong> {employee.email}</div>
                <div><strong>Job Title:</strong> {employee.jobTitle}</div>
                <div><strong>Department:</strong> {employee.department}</div>
                <div><strong>Created At:</strong> {employee.createdAt}</div>
                <div><strong>Modified At:</strong> {employee.modifiedAt}</div>
                <div><strong>Created By:</strong> {employee.createdBy}</div>
                <div><strong>Modified By:</strong> {employee.modifiedBy}</div>
            </div>
            <div className="page-actions">
                <Link to="/">
                    <Button label="Back" outlined />
                </Link>
                <Link to={`/employees/${employee.employeeId}/edit`}>
                    <Button label="Edit" />
                </Link>
                <Button label="Delete" severity="danger" onClick={handleDelete} />
            </div>
        </Card>
    );
}
```

FRONTEND FILE

EmployeeListPage.jsx

employee-portal-frontend/src/pages/EmployeeListPage.jsx

React JSX

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```
import { Card } from "primereact/card";
import DepartmentFilter from "../component/DepartmentFilter";
import { useEffect, useState } from "react";
import { Message } from 'primereact/message';
import { ProgressSpinner } from "primereact/progressspinner";
import { ConfirmDialog, confirmDialog } from "primereact/confirmdialog";
import { EmployeeService } from "../api/employeeApi";
import EmployeeTable from "../component/EmployeeTable";

export default function EmployeeListPage() {

    const [department, setDepartment] = useState("HR");
    const [employees, setEmployees] = useState([]);
    const [loading, setIsLoading] = useState(false);
    const [error, setError] = useState("");

    useEffect(() => {
        const fetchData = async () => {
            setIsLoading(true);
            setError("");
            try {
                const data = await EmployeeService.getEmployeesByDepartment(department);
                setEmployees(data);
            } catch (err) {
                console.log(err);
                setEmployees([]);
                setError("Err....We ran into some trouble fetching data");
            } finally {
                setIsLoading(false);
            }
        };

        fetchData();
    }, [department])

    const handleDepartmentChange = async (dept) => {
        setDepartment(dept);
    }

    const handleDelete = (id) => {
        confirmDialog({
            message: "Last Chance!!! Are you sure to delete ?",
            header: "Delete Employee",
            icon: "pi pi-exclamation-triangle",
```

```
            acceptClassName: "p-button-danger",
            accept: async () => {
                try {
                    setError('');
                    await EmployeeService.deleteEmployee(id);
                    const data = await EmployeeService.getEmployeesByDepartment(department);
                    setEmployees(data);
                } catch (err) {
                    setError('Failed to delete employee.');
                }
            },
        });
    }

    return (
        <Card title="Employee List" className="page-card">
            <ConfirmDialog />
            <DepartmentFilter
                department={department}
                onDepartmentChange={handleDepartmentChange}
            />

            {!department && !loading && (
                <p className="muted-text">Choose a department to view employees.</p>
            )}

            {loading && (
                <div className="centered-block">
                    <ProgressSpinner />
                </div>
            )}

            {error && (
                <div className="message-block">
                    <Message severity="error" text={error} />
                </div>
            )}

            {!loading && !error && department && (
                <EmployeeTable employees={employees} onDelete={handleDelete} />
            )}
        </Card>
    );
}
```

FRONTEND FILE

NotFoundPage.jsx

employee-portal-frontend/src/pages/NotFoundPage.jsx

React JSX

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```
import { Link } from 'react-router-dom';
import { Card } from 'primereact/card';
import { Button } from 'primereact/button';

export default function NotFoundPage() {
    return (
        <Card title={"404 - Page Not Found"} className="page-card">
            <p>The page you are looking for does not exist.</p>
            <Link to="/">
                <Button label={"Go Home"} />
            </Link>
        </Card>
    );
}
```

SECTION 2

Backend

This section contains the Spring Boot backend files, including application setup, REST endpoints, DTOs, business logic, persistence, exception handling, configuration, migrations, and the current test class.

```
https://github.com/kezama1231/EmployeePortalApp
```

BACKEND FILE

pom.xml

EmployeePortalApp/pom.xml

XML

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```xml
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-
4.0.0.xsd">
        <modelVersion>4.0.0</modelVersion>
        <parent>
                <groupId>org.springframework.boot</groupId>
                <artifactId>spring-boot-starter-parent</artifactId>
                <version>3.5.12-SNAPSHOT</version>
                <relativePath/> <!-- lookup parent from repository -->
        </parent>
        <groupId>com.interview</groupId>
        <artifactId>EmployeePortalApp</artifactId>
        <version>0.0.1-SNAPSHOT</version>
        <name>EmployeePortalApp</name>
        <description>Demo MVP project for Spring Boot &amp; React</description>
        <url/>
        <licenses>
                <license/>
        </licenses>
        <developers>
                <developer/>
        </developers>
        <scm>
                <connection/>
                <developerConnection/>
                <tag/>
                <url/>
        </scm>
        <properties>
                <java.version>17</java.version>
                <org.mapstruct.version>1.6.3</org.mapstruct.version>
                <lombok.version>1.18.38</lombok.version>
        </properties>
        <dependencies>
                <dependency>
                        <groupId>org.springframework.boot</groupId>
                        <artifactId>spring-boot-starter-data-jpa</artifactId>
                </dependency>
                <dependency>
                        <groupId>org.springframework.boot</groupId>
                        <artifactId>spring-boot-starter-validation</artifactId>
                </dependency>
                <dependency>
```

```
                <groupId>org.springframework.boot</groupId>
                <artifactId>spring-boot-starter-web</artifactId>
        </dependency>
        <dependency>
                <groupId>org.flywaydb</groupId>
                <artifactId>flyway-core</artifactId>
        </dependency>

        <dependency>
                <groupId>com.h2database</groupId>
                <artifactId>h2</artifactId>
                <scope>runtime</scope>
        </dependency>
        <dependency>
                <groupId>org.projectlombok</groupId>
                <artifactId>lombok</artifactId>
                <optional>true</optional>
        </dependency>
        <dependency>
                <groupId>org.springframework.boot</groupId>
                <artifactId>spring-boot-starter-test</artifactId>
                <scope>test</scope>
        </dependency>
        <dependency>
                <groupId>org.mapstruct</groupId>
                <artifactId>mapstruct</artifactId>
                <version>${org.mapstruct.version}</version>
        </dependency>
</dependencies>

<build>
        <plugins>
                <plugin>
                        <groupId>org.apache.maven.plugins</groupId>
                        <artifactId>maven-compiler-plugin</artifactId>
                        <configuration>
                                <annotationProcessorPaths>
                                        <path>
                                                <groupId>org.projectlombok</groupId>
                                                <artifactId>lombok</artifactId>
                                                <version>${lombok.version}</version>
                                        </path>
                                        <path>
                                                <groupId>org.projectlombok</groupId>
                                                <artifactId>lombok-mapstruct-binding</artifactId>
                                                <version>0.2.0</version>
                                        </path>
                                        <path>
                                                <groupId>org.mapstruct</groupId>
                                                <artifactId>mapstruct-processor</artifactId>
                                                <version>${org.mapstruct.version}</version>
                                        </path>
                                </annotationProcessorPaths>
                        </configuration>
                </plugin>
                <plugin>
                        <groupId>org.springframework.boot</groupId>
                        <artifactId>spring-boot-maven-plugin</artifactId>
                        <configuration>
                                <excludes>
                                        <exclude>
```

```
                                                        <groupId>org.projectlombok</groupId>
                                                        <artifactId>lombok</artifactId>
                                                </exclude>
                                        </excludes>
                                </configuration>
                        </plugin>
                </plugins>
        </build>
        <repositories>
                <repository>
                        <id>spring-snapshots</id>
                        <name>Spring Snapshots</name>
                        <url>https://repo.spring.io/snapshot</url>
                        <releases>
                                <enabled>false</enabled>
                        </releases>
                </repository>
        </repositories>
        <pluginRepositories>
                <pluginRepository>
                        <id>spring-snapshots</id>
                        <name>Spring Snapshots</name>
                        <url>https://repo.spring.io/snapshot</url>
                        <releases>
                                <enabled>false</enabled>
                        </releases>
                </pluginRepository>
        </pluginRepositories>

</project>
```

BACKEND FILE

application.properties

EmployeePortalApp/src/main/resources/application.properties

Properties

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```
spring.application.name=EmployeePortalApp

spring.datasource.url=jdbc:h2:mem:employee_db
spring.datasource.driverClassName=org.h2.Driver
spring.datasource.username=sa
spring.datasource.password=

spring.jpa.hibernate.ddl-auto=validate
spring.jpa.show-sql=true
spring.jpa.properties.hibernate.format_sql=true

spring.h2.console.enabled=true
spring.h2.console.path=/h2-console

spring.flyway.enabled=true
```

BACKEND FILE

V1__create_employee_table.sql

EmployeePortalApp/src/main/resources/db/migration/V1__create_employee_table.sql

SQL

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```sql
CREATE TABLE employee (
    employee_id BIGINT GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    contact_number VARCHAR(20) NOT NULL,
    department VARCHAR(50) NOT NULL,
    email VARCHAR(100) NOT NULL UNIQUE,
    job_title VARCHAR(100) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    modified_at TIMESTAMP NOT NULL,
    created_by VARCHAR(50) NOT NULL,
    modified_by VARCHAR(50) NOT NULL
);
```

BACKEND FILE

V2__insert_employee_seed_data.sql

EmployeePortalApp/src/main/resources/db/migration/V2__insert_employee_seed_data.sql

SQL

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```sql
INSERT INTO employee (
    name,
    contact_number,
    department,
    email,
    job_title,
    created_at,
    modified_at,
    created_by,
    modified_by
) VALUES
    ('Alicia Gomez', '555-0101', 'HR', 'alicia.gomez@company.com', 'HR Manager', TIMESTAMP '2026-03-16 09:00:00', TIMESTAMP '2026-03-16 09:00:00', 'system', 'system'),
    ('Brian Lee', '555-0102', 'HR', 'brian.lee@company.com', 'Recruitment Specialist', TIMESTAMP '2026-03-16 09:05:00', TIMESTAMP '2026-03-16 09:05:00', 'system', 'system'),
    ('Chloe Martin', '555-0103', 'HR', 'chloe.martin@company.com', 'People Operations Analyst', TIMESTAMP '2026-03-16 09:10:00', TIMESTAMP '2026-03-16 09:10:00', 'system', 'system'),
    ('Daniel Wright', '555-0104', 'HR', 'daniel.wright@company.com', 'Compensation Coordinator', TIMESTAMP '2026-03-16 09:15:00', TIMESTAMP '2026-03-16 09:15:00', 'system', 'system'),
    ('Ethan Brooks', '555-0201', 'IT', 'ethan.brooks@company.com', 'Software Engineer', TIMESTAMP '2026-03-16 09:20:00', TIMESTAMP '2026-03-16 09:20:00', 'system', 'system'),
    ('Fatima Khan', '555-0202', 'IT', 'fatima.khan@company.com', 'Frontend Developer', TIMESTAMP '2026-03-16 09:25:00', TIMESTAMP '2026-03-16 09:25:00', 'system', 'system'),
    ('Gabriel Chen', '555-0203', 'IT', 'gabriel.chen@company.com', 'DevOps Engineer', TIMESTAMP '2026-03-16 09:30:00', TIMESTAMP '2026-03-16 09:30:00', 'system', 'system'),
    ('Hannah Patel', '555-0204', 'IT', 'hannah.patel@company.com', 'IT Support Lead', TIMESTAMP '2026-03-16 09:35:00', TIMESTAMP '2026-03-16 09:35:00', 'system', 'system'),
    ('Isaac Turner', '555-0301', 'Finance', 'isaac.turner@company.com', 'Financial Analyst', TIMESTAMP '2026-03-16 09:40:00', TIMESTAMP '2026-03-16 09:40:00', 'system', 'system'),
    ('Jasmine Rivera', '555-0302', 'Finance', 'jasmine.rivera@company.com', 'Accountant', TIMESTAMP '2026-03-16 09:45:00', TIMESTAMP '2026-03-16 09:45:00', 'system', 'system'),
    ('Kevin Murphy', '555-0303', 'Finance', 'kevin.murphy@company.com', 'Payroll Specialist', TIMESTAMP '2026-03-16 09:50:00', TIMESTAMP '2026-03-16 09:50:00', 'system', 'system'),
    ('Lina Hassan', '555-0304', 'Finance', 'lina.hassan@company.com', 'Finance Manager', TIMESTAMP '2026-03-16 09:55:00', TIMESTAMP '2026-03-16 09:55:00', 'system', 'system'),
    ('Marcus Evans', '555-0401', 'Operations', 'marcus.evans@company.com', 'Operations Manager', TIMESTAMP '2026-03-16 10:00:00', TIMESTAMP '2026-03-16 10:00:00', 'system', 'system'),
    ('Nina Silva', '555-0402', 'Operations', 'nina.silva@company.com', 'Logistics Coordinator', TIMESTAMP '2026-03-16 10:05:00', TIMESTAMP '2026-03-16 10:05:00', 'system', 'system'),
    ('Owen Parker', '555-0403', 'Operations', 'owen.parker@company.com', 'Process Improvement Analyst', TIMESTAMP '2026-03-16 10:10:00', TIMESTAMP '2026-03-16 10:10:00', 'system', 'system');
```

BACKEND FILE

EmployeePortalAppApplication.java

EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/EmployeePortalAppApplication.java

Java

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```java
package com.interview.EmployeePortalApp;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class EmployeePortalAppApplication {

	public static void main(String[] args) {
		SpringApplication.run(EmployeePortalAppApplication.class, args);
	}

}
```

BACKEND FILE

AuditConfig.java

EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/config/AuditConfig.java

Java

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```
package com.interview.EmployeePortalApp.config;

import org.springframework.context.annotation.Configuration;
import org.springframework.data.jpa.repository.config.EnableJpaAuditing;

@Configuration
@EnableJpaAuditing(auditorAwareRef="auditorAware")
public class AuditConfig {}
```

BACKEND FILE

AuditorAwareImpl.java

EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/config/AuditorAwareImpl.java

Java

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```java
package com.interview.EmployeePortalApp.config;

import java.util.Optional;

import org.springframework.data.domain.AuditorAware;
import org.springframework.stereotype.Component;

@Component("auditorAware")
public class AuditorAwareImpl implements AuditorAware<String> {

    @Override
    public Optional<String> getCurrentAuditor() {
        return Optional.of("system");
    }
}
```

BACKEND FILE

EmployeeController.java

EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/controller/EmployeeController.java

Java

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```java
package com.interview.EmployeePortalApp.controller;

import java.util.List;

import org.springframework.http.HttpStatus;
import org.springframework.web.bind.annotation.CrossOrigin;
import org.springframework.web.bind.annotation.DeleteMapping;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.PutMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RequestParam;
import org.springframework.web.bind.annotation.ResponseStatus;
import org.springframework.web.bind.annotation.RestController;

import com.interview.EmployeePortalApp.dto.EmployeeCreateRequest;
import com.interview.EmployeePortalApp.dto.EmployeeResponse;
import com.interview.EmployeePortalApp.dto.EmployeeUpdateRequest;
import com.interview.EmployeePortalApp.service.EmployeeService;

import jakarta.validation.Valid;
import lombok.RequiredArgsConstructor;

@RestController
@RequiredArgsConstructor
@RequestMapping("api/employees")
@CrossOrigin(origins = "http://localhost:5173")
public class EmployeeController {

    private final EmployeeService service;

    @PostMapping
    @ResponseStatus(HttpStatus.CREATED)
    public EmployeeResponse createEmployee(@Valid @RequestBody EmployeeCreateRequest request){
        return service.createEmployee(request);
    }

    @GetMapping
    @ResponseStatus(HttpStatus.OK)
    public List<EmployeeResponse> getEmployeeByDepartment(@RequestParam String department){
        return service.getEmployeesByDepartment(department);
    }
```

```
    @GetMapping("/{employeeId}")
    public EmployeeResponse getEmployeeById(@PathVariable Long employeeId){
        return service.getEmployeeById(employeeId);
    }

    @PutMapping("/{employeeId}")
    public EmployeeResponse updateEmployee(
        @PathVariable Long employeeId,
        @RequestBody @Valid EmployeeUpdateRequest request
    ){
        return service.updateEmployee(employeeId, request);
    }

    @DeleteMapping("/{employeeId}")
    @ResponseStatus(HttpStatus.NO_CONTENT)
    public void deleteEmployee(@PathVariable Long employeeId){
        service.deleteEmployee(employeeId);
    }
}
```

BACKEND FILE

EmployeeCreateRequest.java

EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/dto/EmployeeCreateRequest.java

Java

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```java
package com.interview.EmployeePortalApp.dto;

import jakarta.validation.constraints.Email;
import jakarta.validation.constraints.NotBlank;
import jakarta.validation.constraints.Size;
import lombok.Getter;
import lombok.Setter;

@Getter
@Setter
public class EmployeeCreateRequest {

    @NotBlank(message = "Name is required")
    @Size(min = 2, max = 100, message = "Name must be between 2 and 100 characters")
    private String name;

    @NotBlank(message = "Contact number is required")
    @Size(min = 8, max = 20, message = "Contact number must be between 8 and 20 characters")
    private String contactNumber;

    @NotBlank(message = "Department is required")
    @Size(min = 2, max = 50, message = "Department must be between 2 and 50 characters")
    private String department;

    @NotBlank(message = "Email is required")
    @Email(message = "Email format is invalid")
    @Size(max = 100, message = "Email must not exceed 100 characters")
    private String email;

    @NotBlank(message = "Job title is required")
    @Size(min = 2, max = 100, message = "Job title must be between 2 and 100 characters")
    private String jobTitle;

}
```

BACKEND FILE

EmployeeUpdateRequest.java

EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/dto/EmployeeUpdateRequest.java

Java

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```java
package com.interview.EmployeePortalApp.dto;

import jakarta.validation.constraints.Email;
import jakarta.validation.constraints.NotBlank;
import jakarta.validation.constraints.Size;
import lombok.Getter;
import lombok.Setter;

@Getter
@Setter
public class EmployeeUpdateRequest {
    @NotBlank(message = "Name is required")
    @Size(min = 2, max = 100, message = "Name must be between 2 and 100 characters")
    private String name;

    @NotBlank(message = "Contact number is required")
    @Size(min = 8, max = 20, message = "Contact number must be between 8 and 20 characters")
    private String contactNumber;

    @NotBlank(message = "Department is required")
    @Size(min = 2, max = 50, message = "Department must be between 2 and 50 characters")
    private String department;

    @NotBlank(message = "Email is required")
    @Email(message = "Email format is invalid")
    @Size(max = 100, message = "Email must not exceed 100 characters")
    private String email;

    @NotBlank(message = "Job title is required")
    @Size(min = 2, max = 100, message = "Job title must be between 2 and 100 characters")
    private String jobTitle;
}
```

BACKEND FILE

EmployeeResponse.java

EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/dto/EmployeeResponse.java

Java

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```java
package com.interview.EmployeePortalApp.dto;

import java.time.LocalDateTime;

import lombok.Getter;
import lombok.Setter;

@Getter
@Setter
public class EmployeeResponse {
    private Long employeeId;
    private String name;
    private String contactNumber;
    private String department;
    private String email;
    private String jobTitle;
    private LocalDateTime createdAt;
    private LocalDateTime modifiedAt;
    private String createdBy;
    private String modifiedBy;
}
```

BACKEND FILE

Employee.java

EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/entity/Employee.java

Java

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```java
package com.interview.EmployeePortalApp.entity;

import java.time.LocalDateTime;

import org.springframework.data.annotation.CreatedBy;
import org.springframework.data.annotation.CreatedDate;
import org.springframework.data.annotation.LastModifiedBy;
import org.springframework.data.annotation.LastModifiedDate;
import org.springframework.data.jpa.domain.support.AuditingEntityListener;

import jakarta.persistence.Column;
import jakarta.persistence.Entity;
import jakarta.persistence.EntityListeners;
import jakarta.persistence.GeneratedValue;
import jakarta.persistence.GenerationType;
import jakarta.persistence.Id;
import jakarta.persistence.Table;
import lombok.AllArgsConstructor;
import lombok.Builder;
import lombok.Getter;
import lombok.NoArgsConstructor;
import lombok.Setter;


@Entity
@Table(name="employee")
@EntityListeners(AuditingEntityListener.class)
@Getter
@Setter
@NoArgsConstructor
@AllArgsConstructor
@Builder
public class Employee {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long employeeId;

    @Column(nullable = false, length = 100)
    private String name;

    @Column(nullable = false, length = 20)
    private String contactNumber;

    @Column(nullable = false, length = 50)
```

```
    private String department;

    @Column(nullable = false, unique = true, length = 100)
    private String email;

    @Column(nullable = false, length = 100)
    private String jobTitle;

    @CreatedDate
    @Column(nullable = false, updatable = false)
    private LocalDateTime createdAt;

    @CreatedBy
    @Column(nullable = false, updatable = false, length = 50)
    private String createdBy;

    @LastModifiedDate
    @Column(nullable = false)
    private LocalDateTime modifiedAt;

    @LastModifiedBy
    @Column(nullable = false, length = 50)
    private String modifiedBy;
}
```

BACKEND FILE

EmployeeNotFoundException.java

EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/exception/EmployeeNotFoundException.java

Java

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```
package com.interview.EmployeePortalApp.exception;

public class EmployeeNotFoundException extends RuntimeException {
    public EmployeeNotFoundException(Long employeeID){
        super("Employee not found with id: " + employeeID);
    }

}
```

BACKEND FILE

GlobalExceptionHandler.java

EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/exception/GlobalExceptionHandler.java

Java

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```java
package com.interview.EmployeePortalApp.exception;

import java.time.LocalDateTime;
import java.util.HashMap;
import java.util.Map;

import org.springframework.http.HttpStatus;
import org.springframework.web.bind.MethodArgumentNotValidException;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.ResponseStatus;
import org.springframework.web.bind.annotation.RestControllerAdvice;

@RestControllerAdvice
public class GlobalExceptionHandler {

    @ExceptionHandler(EmployeeNotFoundException.class)
    @ResponseStatus(HttpStatus.NOT_FOUND)
    public Map<String, Object> handleEmployeeNoutFound(EmployeeNotFoundException ex){
        Map<String, Object> response = new HashMap<>();
        response.put("timestamp: ", LocalDateTime.now());
        response.put("status :", 404 );
        response.put("error :", "NOT FOUND");
        response.put("message :", ex.getMessage());
        return response;
    }

    @ExceptionHandler(MethodArgumentNotValidException.class)
    @ResponseStatus(HttpStatus.BAD_REQUEST)
    public Map<String, Object> handleRequestValidation (MethodArgumentNotValidException ex){
        Map<String, Object> response = new HashMap<>();
        response.put("timestamp: ", LocalDateTime.now());
        response.put("status :", 400 );
        response.put("error :", "BAD REQUEST");
        response.put("message :", ex.getMessage());
        return response;
    }
}
```

BACKEND FILE

EmployeeMapper.java

EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/mapper/EmployeeMapper.java

Java

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```java
package com.interview.EmployeePortalApp.mapper;

import org.mapstruct.Mapper;
import org.mapstruct.Mapping;
import org.mapstruct.MappingTarget;

import com.interview.EmployeePortalApp.dto.EmployeeCreateRequest;
import com.interview.EmployeePortalApp.dto.EmployeeResponse;
import com.interview.EmployeePortalApp.dto.EmployeeUpdateRequest;
import com.interview.EmployeePortalApp.entity.Employee;

@Mapper(componentModel="spring")
public interface EmployeeMapper {

    @Mapping(target = "employeeId", ignore = true)
    @Mapping(target = "createdAt", ignore = true)
    @Mapping(target = "createdBy", ignore = true)
    @Mapping(target = "modifiedAt", ignore = true)
    @Mapping(target = "modifiedBy", ignore = true)
    Employee toEntity(EmployeeCreateRequest request);


    @Mapping(target = "employeeId", ignore = true)
    @Mapping(target = "createdAt", ignore = true)
    @Mapping(target = "createdBy", ignore = true)
    @Mapping(target = "modifiedAt", ignore = true)
    @Mapping(target = "modifiedBy", ignore = true)
    void updateEntityFromRequest(EmployeeUpdateRequest request, @MappingTarget Employee employee);

    EmployeeResponse toResponse(Employee employee);
}
```

BACKEND FILE

EmployeeRepository.java

EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/repository/EmployeeRepository.java

Java

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```java
package com.interview.EmployeePortalApp.repository;

import java.util.List;

import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.stereotype.Repository;

import com.interview.EmployeePortalApp.entity.Employee;

@Repository
public interface EmployeeRepository extends JpaRepository<Employee, Long> {
    List<Employee> findByDepartmentOrderByNameAsc(String department);
}
```

BACKEND FILE

EmployeeService.java

EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/service/EmployeeService.java

Java

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```java
package com.interview.EmployeePortalApp.service;

import java.util.List;

import com.interview.EmployeePortalApp.dto.EmployeeCreateRequest;
import com.interview.EmployeePortalApp.dto.EmployeeResponse;
import com.interview.EmployeePortalApp.dto.EmployeeUpdateRequest;

public interface EmployeeService {

    EmployeeResponse createEmployee(EmployeeCreateRequest request);

    List<EmployeeResponse> getEmployeesByDepartment(String department);

    EmployeeResponse getEmployeeById(Long employeeId);

    EmployeeResponse updateEmployee(Long employeeId, EmployeeUpdateRequest request);

    EmployeeResponse deleteEmployee(Long employeeId);

}
```

BACKEND FILE

EmployeeServiceImpl.java

EmployeePortalApp/src/main/java/com/interview/EmployeePortalApp/service/impl/EmployeeServiceImpl.java

Java

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```java
package com.interview.EmployeePortalApp.service.impl;

import java.util.List;

import org.springframework.stereotype.Service;

import com.interview.EmployeePortalApp.dto.EmployeeCreateRequest;
import com.interview.EmployeePortalApp.dto.EmployeeResponse;
import com.interview.EmployeePortalApp.dto.EmployeeUpdateRequest;
import com.interview.EmployeePortalApp.entity.Employee;
import com.interview.EmployeePortalApp.exception.EmployeeNotFoundException;
import com.interview.EmployeePortalApp.mapper.EmployeeMapper;
import com.interview.EmployeePortalApp.repository.EmployeeRepository;
import com.interview.EmployeePortalApp.service.EmployeeService;

import lombok.RequiredArgsConstructor;

@Service
@RequiredArgsConstructor
public class EmployeeServiceImpl implements EmployeeService {

    private final EmployeeRepository repository;
    private final EmployeeMapper mapper;

    @Override
    public EmployeeResponse createEmployee(EmployeeCreateRequest request){
        Employee employee = mapper.toEntity(request);
        Employee savedEmployee = repository.save(employee);
        return mapper.toResponse(savedEmployee);
    }

    @Override
    public List<EmployeeResponse> getEmployeesByDepartment(String department){
        List<EmployeeResponse> list = repository.findByDepartmentOrderByNameAsc(department)
                                        .stream()
                                        .map(mapper::toResponse)
                                        .toList();
        return list;

    }

    @Override
    public EmployeeResponse getEmployeeById(Long employeeId){
        Employee emp = repository.findById(employeeId)
```

```
                        .orElseThrow(() -> new EmployeeNotFoundException(employeeId));
        return mapper.toResponse(emp);

    }

    @Override
    public EmployeeResponse updateEmployee(Long employeeId, EmployeeUpdateRequest request){
        Employee emp = repository.findById(employeeId)
                        .orElseThrow(() -> new EmployeeNotFoundException(employeeId));
        mapper.updateEntityFromRequest(request, emp);
        return mapper.toResponse(repository.save(emp));
    }

    @Override
    public EmployeeResponse deleteEmployee(Long employeeId){
        Employee emp = repository.findById(employeeId)
                        .orElseThrow(() -> new EmployeeNotFoundException(employeeId));
        repository.delete(emp);
        return mapper.toResponse(emp);
    }
}
```

BACKEND FILE

EmployeePortalAppApplicationTests.java

EmployeePortalApp/src/test/java/com/interview/EmployeePortalApp/EmployeePortalAppApplicationTests.java

Java

GITHUB REPO

https://github.com/kezama1231/EmployeePortalApp

```
package com.interview.EmployeePortalApp;

import org.junit.jupiter.api.Test;
import org.springframework.boot.test.context.SpringBootTest;

@SpringBootTest
class EmployeePortalAppApplicationTests {

        @Test
        void contextLoads() {
        }

}
```